Use Case Diagram and Specifications

Digital Document Verification System

Submitted By:
Ansh Tyagi
Roll No: 202401100200058
Branch: CSE
Section: A

Subject:
Software Engineering and Design Principles

Contents

1 Use Case Diagram and Specifications **2**
- 1.1 Use Case Diagram . . . 2
- 1.2 Actors and Their Roles . . . 3
- 1.3 Use Case Specifications . . . 4
  - 1.3.1 UC-1: Register / Login . . . 4
  - 1.3.2 UC-2: Upload Document . . . 4
  - 1.3.3 UC-3: Verify QR Code . . . 4
  - 1.3.4 UC-4: Verify Digital Signature . . . 4
  - 1.3.5 UC-5: Verify Document Data . . . 4
  - 1.3.6 UC-6: View Verification Result . . . 5
  - 1.3.7 UC-7: View Verification History . . . 5
  - 1.3.8 UC-8: Generate Report . . . 5
  - 1.3.9 UC-9: Manage Users and Roles . . . 5
  - 1.3.10 UC-10: Configure Verification Rules . . . 5
  - 1.3.11 UC-11: Review Audit Logs . . . 6
  - 1.3.12 UC-12: Provide Document Records . . . 6
  - 1.3.13 UC-13: Update Document Status . . . 6
  - 1.3.14 UC-14: Receive Notifications . . . 6
  - 1.3.15 UC-15: Maintain System . . . 6
- 1.4 Use Case Summary Table . . . 7

Chapter 1

Use Case Diagram and Specifications

1.1 Use Case Diagram

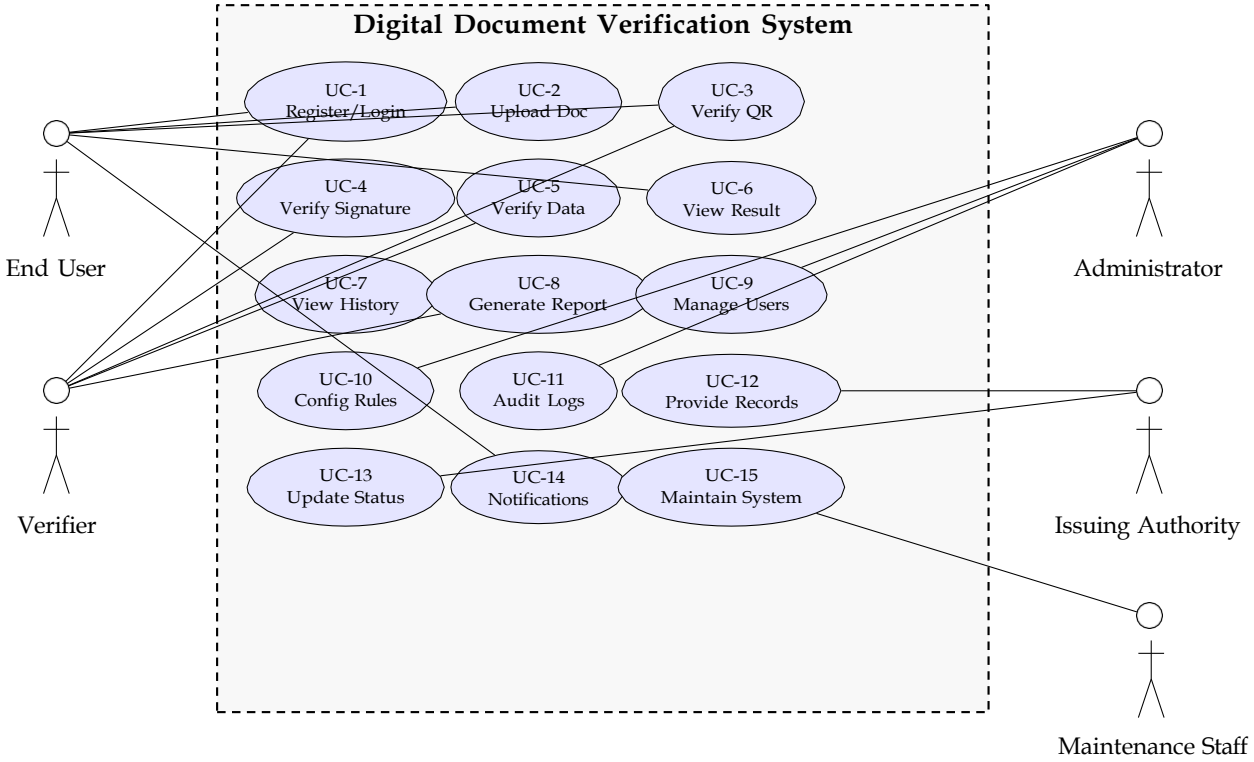


Figure 1.1: Use Case Diagram — Digital Document Verification System

1.2 Actors and Their Roles

Actor	Role
End User / Document Holder	Normal user who wants to verify the authenticity of a digital document. Can register, login, upload documents, scan QR codes, view results, and view history. Cannot access administrative functions.
Verifier / Authorized Organization User	Authorized employee or organization representative who verifies documents submitted by others. Has all End User capabilities plus can verify multiple documents and view detailed results based on permissions.
Administrator	Manages the overall system, users, roles, verification rules, and audit records. Has highest privileges. Monitors system activities and handles verification issues.
Issuing Authority	Organization (university, government, company) that originally creates and issues documents. Maintains official records, provides verification data via API/database, digitally signs documents, and can update document status.
System Maintenance / Technical Staff	Responsible for maintaining, monitoring, updating, and troubleshooting the system. Has advanced technical knowledge of servers, databases, APIs, and security. Access restricted per security policy.

1.3 Use Case Specifications

1.3.1 UC-1: Register / Login

Actor(s): All actors
Precondition: User has internet access and a modern web browser. User is not logged in.
Postcondition: User is successfully registered/logged in and granted access based on role.
Function: Allows new users to register with valid email and password, and existing users to log in with credentials. Implements role-based access control (RBAC).

1.3.2 UC-2: Upload Document

Actor(s): End User, Verifier
Precondition: User is logged in. Document is available in supported format (PDF, JPEG, PNG, TIFF).
Postcondition: Document is uploaded securely, and processing begins.
Function: Allows users to upload documents via drag-and-drop or file browser. Validates file type and size, then extracts text, QR codes, barcodes, and digital signatures.

1.3.3 UC-3: Verify QR Code

Actor(s): End User, Verifier
Precondition: User is logged in. QR code is present on the document or available as an image. Device camera is available (for scanning).
Postcondition: QR code is decoded and validated against trusted records; result is displayed.
Function: Scans QR codes using device camera or uploaded image, decodes embedded information, and validates it against trusted system records or external APIs.

1.3.4 UC-4: Verify Digital Signature

Actor(s): End User, Verifier
Precondition: User is logged in. Document contains a digital signature. Issuer's public key/certificate is available from a trusted source.
Postcondition: Digital signature is verified; tampering is detected if present; result is displayed.
Function: Detects and extracts digital signatures from documents, verifies them using trusted public keys/certificates, and detects tampering by comparing document hashes.

1.3.5 UC-5: Verify Document Data

Actor(s): End User, Verifier
Precondition: User is logged in. Document has been uploaded and processed. Trusted database or API is accessible.
Postcondition: Extracted data is compared with trusted records; match/mismatch result is displayed.

Function: Extracts relevant data fields (name, ID number, issue date) from the document, queries trusted databases/APIs, and compares extracted data with trusted records.

1.3.6 UC-6: View Verification Result

Actor(s): End User, Verifier
Precondition: User is logged in. At least one verification has been completed.
Postcondition: Verification result (Verified / Invalid / Manual Review Required) is displayed with details.
Function: Displays the final verification result after all checks complete, including details of what passed/failed.

1.3.7 UC-7: View Verification History

Actor(s): End User, Verifier
Precondition: User is logged in. At least one verification has been performed previously.
Postcondition: List of past verification activities is displayed with filters and search options.
Function: Maintains and displays records of previous verification activities, allowing users to search and filter history.

1.3.8 UC-8: Generate Report

Actor(s): End User, Verifier
Precondition: User is logged in. Verification has been completed.
Postcondition: Verification report is generated in PDF format and available for download.
Function: Generates a downloadable PDF report containing verification results, timestamps, and details.

1.3.9 UC-9: Manage Users and Roles

Actor(s): Administrator
Precondition: Administrator is logged in with administrative privileges.
Postcondition: Users are created, edited, deleted, or assigned roles; changes are saved.
Function: Allows administrators to create, edit, delete users and assign roles (End User, Verifier, Admin, etc.). Implements RBAC.

1.3.10 UC-10: Configure Verification Rules

Actor(s): Administrator
Precondition: Administrator is logged in.
Postcondition: Verification rules are configured and saved; applied to future verifications.
Function: Allows administrators to configure verification rules, thresholds, and system settings for document validation.

1.3.11 UC-11: Review Audit Logs

Actor(s): Administrator
Precondition: Administrator is logged in. Audit logs exist in the system.
Postcondition: Audit logs are displayed for review; suspicious activities can be identified.
Function: Displays audit logs of all user activities and system events for security monitoring and compliance.

1.3.12 UC-12: Provide Document Records

Actor(s): Issuing Authority
Precondition: Issuing authority is authorized and has access to the system's API/interface.
Postcondition: Document records are provided to the system for verification purposes.
Function: Allows issuing authorities to provide official document records through a secure API or database interface for verification.

1.3.13 UC-13: Update Document Status

Actor(s): Issuing Authority
Precondition: Issuing authority is authorized. Document record exists in the system.
Postcondition: Document status is updated (valid, expired, revoked) in the system.
Function: Allows issuing authorities to update the status of documents they originally issued (e.g., mark as expired or revoked).

1.3.14 UC-14: Receive Notifications

Actor(s): All actors
Precondition: User is registered. Notification preferences are configured.
Postcondition: User receives notification via email or in-app notification about verification results or required actions.
Function: Sends email and in-app notifications to users about verification results, manual review requirements, and other important events.

1.3.15 UC-15: Maintain System

Actor(s): System Maintenance / Technical Staff
Precondition: Technical staff has authorized access. System is running.
Postcondition: System is maintained, monitored, updated, and troubleshooted as needed.
Function: Performs system maintenance, monitoring, updates, backups, and troubleshooting to ensure smooth operation.

1.4 Use Case Summary Table

UC ID	Use Case	Primary Actor	Precondition	Postcondition
UC-1	Register / Login	All	Not logged in	Logged in with role
UC-2	Upload Document	End User, Verifier	Logged in, valid file	Uploaded and processed
UC-3	Verify QR Code	End User, Verifier	Logged in, QR present	QR validated
UC-4	Verify Digital Signature	End User, Verifier	Logged in, signature present	Signature verified
UC-5	Verify Document Data	End User, Verifier	Logged in, DB/API accessible	Data compared
UC-6	View Verification Result	End User, Verifier	Verification completed	Result displayed
UC-7	View Verification History	End User, Verifier	Past verifications exist	History displayed
UC-8	Generate Report	End User, Verifier	Verification completed	PDF report ready
UC-9	Manage Users and Roles	Administrator	Admin logged in	Users managed
UC-10	Configure Verification Rules	Administrator	Admin logged in	Rules configured
UC-11	Review Audit Logs	Administrator	Admin logged in	Logs displayed
UC-12	Provide Document Records	Issuing Authority	Authorized access	Records provided
UC-13	Update Document Status	Issuing Authority	Authorized, record exists	Status updated
UC-14	Receive Notifications	All	Registered, preferences set	Notification received
UC-15	Maintain System	Maintenance Staff	Authorized access	System maintained